

FUNCTIONAL SAFETY REQUIREMENTS SPECIFICATION (FSRS)

Airbag Electronic Control Unit (ECU)

Functional Safety Assessment per ISO 26262 (ASIL) & IEC 61508 (SIL)

1. Document Information

Item	Value
Document Number	FSRS-AIRBAG-ECU-001
Revision	2.1
Date	2026-05-11
System	Airbag Electronic Control Unit (ECU)
Applicable Standards	ISO 26262:2018 / IEC 61508:2010
Classification	Confidential Functional Safety Document
Author	Functional Safety Engineering Team
Based On	Hazard Analysis & Risk Assessment (HARA)

2. Purpose and Scope

This Functional Safety Requirements Specification (FSRS) defines the functional safety requirements for the Airbag Electronic Control Unit (ECU) laboratory implementation derived from the Hazard Analysis and Risk Assessment (HARA).

The purpose of this document is to translate identified hazards, safety goals, ASIL classifications, and SIL targets into concrete, verifiable, and testable safety requirements applicable to the ECU hardware and embedded software implementation.

The Airbag ECU is a safety-critical automotive electronic system responsible for:

- Monitoring crash sensor signals continuously
- Detecting frontal, lateral, rollover, and impact events
- Making deterministic deployment decisions
- Preventing unintended airbag deployment
- Suppressing passenger airbag deployment during child-seat occupancy
- Performing continuous diagnostics and fault monitoring
- Broadcasting deployment and diagnostic information via CAN communication

This specification applies to the confirmed laboratory architecture consisting of:

- Two Arduino Due boards operating in a redundant 2oo2 safety architecture
- Three potentiometer-based analog crash sensor simulations
- One digital baby-seat switch input
- CAN bus communication to the OMS laboratory PC
- CRC-protected CAN communication
- Parallel CAN bitstream readback verification

The physical pyrotechnic inflator, airbag cushion, and vehicle body structure are treated as adjacent systems outside the laboratory implementation boundary.

3. System Overview

3.1 System Description

The Airbag ECU continuously monitors crash-related sensor information and determines whether airbag deployment is required based on sensor fusion algorithms and occupant classification logic.

The ECU implementation includes:

- Accelerometer simulation
- Gyroscope/angular rate simulation
- Pressure sensor simulation
- Occupant classification (baby-seat detection)
- Redundant processing channels
- CAN communication interface
- Diagnostic and fault monitoring

The system is safety-critical because both unintended deployment and non-deployment during a crash can directly result in severe injury or fatality.

3.2 System Architecture

The laboratory implementation uses a dual-channel redundant architecture (2oo2 — two-out-of-two voting logic).

Two Arduino Due boards independently:

- Read all sensor inputs
- Execute crash detection algorithms
- Calculate deployment decisions
- Generate CRC-protected CAN frames
- Verify communication integrity

A deployment command is issued only if both channels independently reach identical deployment decisions.

This redundancy architecture is required to satisfy:

- ISO 26262 ASIL D
- IEC 61508 SIL 3

ASIL Rationale

A single Arduino Due processing channel does not satisfy the ASIL D single-point fault metric (SPFM) targets required by ISO 26262. Therefore, two independent processing channels operating in a 2oo2 redundant architecture are required to achieve the necessary fault tolerance, decision integrity, and prevention of unintended deployment caused by single-point failures.

3.3 Sensor Inputs (Breadboard)

#	Real Sensor	Laboratory Substitute	Signal Type	Voltage Range	Represents
1	Accelerometer	POTI-1	Analog (ADC)	0–3.3 V	0–100 g crash pulse
2	Gyroscope / Angular Rate	POTI-2	Analog (ADC)	0–3.3 V	0–360°/s rollover / yaw
3	Pressure Sensor	POTI-3	Analog (ADC)	0–3.3 V	0–500 kPa impact pressure
4	Baby Seat / OCS	Digital Switch	Digital GPIO	LOW/HIGH	Passenger airbag suppression

LOW = baby seat detected

HIGH = adult occupant / empty seat

3.4 Output Interfaces

Interface	Description
CAN Bus	Deployment and diagnostic communication
CRC Field	CAN communication integrity verification
Parallel Readback Channel	Bitstream verification
OMS Laboratory PC	Monitoring and logging interface

The deployment decision and sensor status are transmitted over CAN bus to the OMS laboratory PC, where messages are logged and verified.

4. Safety Goals

Safety Goal ID	Safety Goal	ASIL	SIL
SG-01	The ECU shall not deploy airbags without a valid crash event	ASIL D	SIL 3
SG-02	The ECU shall deploy airbags within the required timing window during valid crash events	ASIL D	SIL 3
SG-03	The ECU shall suppress passenger airbag deployment when a child seat is detected	ASIL D	SIL 3
SG-04	The ECU shall activate pretensioners within required timing limits	ASIL C	SIL 2
SG-05	The ECU shall reject EMC disturbances and invalid electrical events	ASIL C	SIL 2

SG-06	The ECU shall detect and report safety-critical faults	ASIL A	SIL 1
SG-07	The ECU shall transmit crash notification CAN frames reliably	ASIL B	SIL 2

5. Functional Safety Requirements

5.1 Crash Detection and Deployment Requirements

FSR-01 — Crash Detection Timing

Requirement

The Airbag ECU shall detect crash conditions within a maximum of 20 ms after crash sensor thresholds are exceeded.

Safety Rationale

Delayed crash detection reduces occupant protection effectiveness and may result in occupant contact with vehicle structures before airbag inflation.

ASIL / SIL

ASIL D / SIL 3

Source Hazard

H2-01, H2-02

FSR-02 — Crash Algorithm Cycle Time

Requirement

The crash detection algorithm shall execute with a deterministic cycle time not exceeding 1 ms.

Safety Rationale

Deterministic execution is required to ensure predictable deployment timing behavior.

ASIL / SIL

ASIL D / SIL 3

Source Hazard

H2-04

FSR-03 — Deployment Authorization Logic

Requirement

The ECU shall authorize airbag deployment only when:

1. Crash thresholds are exceeded
2. Channel A independently detects a valid crash

3. Channel B independently detects a valid crash
4. Both channels agree on the deployment decision
5. No critical fault condition exists
6. Passenger suppression conditions are not active

Safety Rationale

All six deployment authorization conditions are required to prevent unintended deployment caused by sensor faults, software faults, communication corruption, or single-channel processing failures. Independent crash confirmation by both redundant channels combined with fault-free operating conditions ensures that deployment occurs only during verified crash events and prevents single-point failures from causing hazardous airbag activation.

ASIL / SIL

ASIL D / SIL 3

Source Hazard

H1-01, H2-01

FSR-04 — Prevention of Unintended Deployment

Requirement

The ECU shall inhibit all deployment outputs whenever crash thresholds are not exceeded.

Safety Rationale

Unintended deployment during normal driving represents the highest-severity hazard identified in the HARA.

ASIL / SIL

ASIL D / SIL 3

Source Hazard

H1-01, H1-02

FSR-05 — Deployment Command Timing

Requirement

The ECU shall transmit the deployment command within 40 ms after crash detection.

Safety Rationale

The 40 ms deployment command transmission limit is derived from occupant protection timing requirements and representative frontal crash pulse characteristics. Delayed deployment beyond this interval may significantly reduce airbag effectiveness and increase occupant injury risk.

ASIL / SIL

ASIL D / SIL 3

Source Hazard

H2-01

FSR-06 — Side-Airbag Timing Constraint

Requirement

Side-impact deployment logic shall complete within 8 ms after side-impact threshold detection.

Safety Rationale

Side-impact crash events develop significantly faster than frontal impacts due to reduced crush space between the occupant and intrusion zone. The 8 ms maximum deployment timing requirement ensures sufficient occupant protection before lateral intrusion reaches the occupant compartment.

ASIL / SIL

ASIL D / SIL 3

Source Hazard

H2-02

5.2 Redundancy and Decision Integrity Requirements

FSR-07 — Dual-Channel Redundancy

Requirement

Both Arduino Due channels shall independently:

- Acquire sensor data
- Execute crash algorithms
- Calculate deployment decisions
- Generate CAN communication frames
- Calculate CRC values

Safety Rationale

Dual-channel redundancy is required to satisfy ASIL D fault tolerance objectives and reduce the probability of unsafe deployment decisions caused by hardware faults, software faults, timing failures, or corrupted sensor processing within a single processing channel.

ASIL / SIL

ASIL D / SIL 3

FSR-08 — Channel Agreement Verification

Requirement

The ECU shall issue deployment outputs only if both redundant channels independently generate identical deployment decisions.

Safety Rationale

This requirement prevents single-point failures from causing unintended deployment.

ASIL / SIL

ASIL D / SIL 3

5.3 Occupant Classification Requirements

FSR-09 — Passenger Airbag Suppression

Requirement

The ECU shall suppress passenger airbag deployment whenever the baby-seat signal is LOW.

ASIL / SIL

ASIL D / SIL 3

Source Hazard

H1-04

FSR-10 — Suppress-on-Fault Strategy

Requirement

If the baby-seat signal becomes invalid, intermittent, unstable, or unavailable, the ECU shall default to passenger airbag suppression.

Safety Rationale

Suppressing deployment during uncertain occupancy conditions minimizes risk to child occupants.

ASIL / SIL

ASIL D / SIL 3

Source Hazard

H1-04

FSR-11 — Digital Input Debounce

Requirement

The baby-seat digital input shall remain stable for at least 10 ms before a state transition is accepted as valid.

ASIL / SIL

ASIL D / SIL 3

Source Hazard

Intermittent/noisy signal condition

FSR-12 — Baby Seat Input Line Monitoring

Requirement

The ECU shall periodically verify continuity of the baby-seat digital input line.

ASIL / SIL

ASIL D / SIL 3

Source Hazard

Open-circuit and stuck-HIGH faults

5.4 CAN Communication Safety Requirements

FSR-13 — CRC-Protected CAN Communication

Requirement

All CAN frames shall include CRC protection calculated using the Arduino CRC library.

ASIL / SIL

ASIL B / SIL 2

FSR-14 — CAN Bitstream Readback Verification

Requirement

The ECU shall implement a secondary parallel readback channel to verify transmitted CAN bitstreams.

Safety Rationale

Parallel verification detects communication corruption and CAN transmission faults.

ASIL / SIL

ASIL B / SIL 2

FSR-15 — CAN Communication Timeout Detection

Requirement

The ECU shall detect CAN communication timeout conditions exceeding 50 ms.

ASIL / SIL

ASIL B / SIL 2

FSR-16 — Crash Notification Transmission

Requirement

The ECU shall transmit a crash notification CAN frame within 15 ms after deployment authorization.

ASIL / SIL

ASIL B / SIL 2

Source Hazard

H4-02

FSR-16A — Safety-Critical Deployment Communication Integrity**Requirement**

Although the CAN communication channel is classified ASIL B / SIL 2 for general diagnostic communication, deployment-related CAN frames shall achieve end-to-end ASIL D integrity through redundant channel agreement verification, CRC protection, parallel bitstream readback validation, timeout supervision, and deployment authorization cross-checking between both processing channels.

Safety Rationale

In the laboratory implementation, deployment authorization information is transmitted over CAN communication. Additional end-to-end integrity mechanisms are therefore required to ensure that ASIL D deployment decisions are not degraded by the lower inherent integrity classification of the CAN transport layer.

ASIL / SIL

ASIL D / SIL 3

Source Hazard

H1-01, H2-01, H4-02

5.5 Diagnostic and Fault Handling Requirements**FSR-17 — Power-On Self-Test (POST)****Requirement**

The ECU shall execute a complete power-on self-test within 800 ms after startup.

The POST shall verify:

- ADC functionality
- Sensor plausibility
- CAN communication
- CRC integrity
- Memory integrity
- Clock integrity
- Digital input continuity

ASIL / SIL

ASIL C / SIL 2

FSR-18 — Continuous Diagnostic Monitoring

Requirement

The ECU shall execute continuous background diagnostics with intervals not exceeding 100 ms.

ASIL / SIL

ASIL C / SIL 2

FSR-19 — Diagnostic Warning Indicator

Requirement

The ECU shall activate the Diagnostic Warning Indicator (DWI) within 800 ms after detection of a safety-critical fault.

ASIL / SIL

ASIL A / SIL 1

FSR-20 — Safe-State Transition

Requirement

Upon detection of an unrecoverable safety-critical fault, the ECU shall transition into a safe inhibit state in which all deployment outputs are disabled.

ASIL / SIL

ASIL D / SIL 3

FSR-21 — Fault Logging

Requirement

Detected faults shall be stored as Diagnostic Trouble Codes (DTCs).

ASIL / SIL

ASIL A / SIL 1

FSR-22 — Memory Integrity Protection

Requirement

The ECU software shall implement memory integrity protection mechanisms including:

- RAM integrity verification using CRC or equivalent techniques
- Stack boundary monitoring
- Detection of invalid memory access conditions
- Watchdog supervision for runaway software execution
- Safe-state transition upon detection of unrecoverable memory faults

Safety Rationale

Memory corruption, stack overflow, or uncontrolled software execution may lead to loss of deployment control, incorrect deployment authorization, or failure to enter a safe inhibit state. Memory integrity supervision is therefore required to maintain deterministic and safe ECU operation.

ASIL / SIL

ASIL D / SIL 3

Source Hazard

H1-01, H2-01, H4-01

6. Safety Constraints

Constraint ID	Safety Constraint
SC-01	Overall system classification shall remain ASIL D / SIL 3
SC-02	PMHF shall remain below 10^{-8} failures/hour
SC-03	No single-point fault shall cause unintended deployment
SC-04	Deployment requires agreement of both redundant channels
SC-05	CAN communication shall include CRC verification
SC-06	Critical faults shall force transition to a safe inhibit state
SC-07	Safety-related software shall execute deterministically
SC-08	Diagnostic monitoring shall execute continuously during operation
SC-09	Deployment inhibit functionality shall remain enforceable by hardware independent of software execution

7. Timing Requirements

Function	Maximum Allowed Time
Crash detection cycle	1 ms
Deployment decision	10–15 ms
Crash detection response	20 ms
Deployment command transmission	40 ms
Side-impact deployment	< 8 ms

<i>Pretensioner activation</i>	< 5 ms
<i>CAN timeout detection</i>	50 ms
<i>Diagnostic cycle interval</i>	100 ms
<i>POST execution</i>	800 ms
<i>Baby-seat debounce</i>	10 ms

8. Safe States Definition

Safe State ID	Description	System Condition
SS-01	Deployment Inhibit State	All firing outputs disabled
SS-02	Deployment Armed State	Crash monitoring active
SS-03	Passenger Airbag Suppressed	Passenger airbag disabled
SS-04	Partial Degradation State	Faulted channels isolated
SS-05	System Shutdown State	Hardware safety inhibit active and deployment outputs forced LOW independent of software state

9. Interface Requirements

9.1 Sensor Interfaces

Interface	Type	Voltage Range
<i>POTI-1 Accelerometer</i>	Analog ADC	0–3.3 V
<i>POTI-2 Gyroscope</i>	Analog ADC	0–3.3 V
<i>POTI-3 Pressure Sensor</i>	Analog ADC	0–3.3 V
<i>Baby Seat Switch</i>	Digital GPIO	LOW/HIGH

9.2 Processing Interfaces

Interface	Description
<i>Channel A</i>	Arduino Due #1
<i>Channel B</i>	Arduino Due #2
<i>Cross-Comparison Interface</i>	Redundant decision synchronization

9.3 Communication Interfaces

Interface	Description
CAN Bus	Deployment and diagnostic communication
CRC Verification	Frame integrity verification
Parallel Readback	Bitstream validation
OMS Laboratory PC	Logging and monitoring interface

CAN Communication Integrity Note

Deployment-related communication transmitted over the CAN interface shall use CRC protection, timeout monitoring, redundant channel agreement verification, and parallel readback validation to maintain end-to-end ASIL D deployment integrity.

9.4 CAN Deployment Frame Definition

Field / Bit	Description
Bit 0	Driver airbag deployment command
Bit 1	Passenger airbag deployment command
Bit 2	Passenger suppression active
Bit 3	Crash event detected
Bit 4	Channel A fault detected
Bit 5	Channel B fault detected
Bit 6	Diagnostic warning indicator active
Bit 7	Safe-state inhibit active
Byte 1	Deployment sequence counter
Byte 2–3	Sensor status information
Byte 4–5	Diagnostic status information
Byte 6–7	CRC field

CRC Definition

CRC protection shall use a 16-bit CRC algorithm implemented consistently on both processing channels and verified during CAN frame reception and parallel readback validation.

10. Requirement Traceability Matrix

<i>Requirement ID</i>	<i>Safety Goal</i>	<i>Governing Hazard</i>	<i>ASIL</i>	<i>SIL</i>
<i>FSR-01</i>	SG-02	H2-01	D	3
<i>FSR-02</i>	SG-02	H2-04	D	3
<i>FSR-03</i>	SG-01	H1-01	D	3
<i>FSR-04</i>	SG-01	H1-01	D	3
<i>FSR-05</i>	SG-02	H2-01	D	3
<i>FSR-06</i>	SG-02	H2-02	D	3
<i>FSR-07</i>	SG-01	H1-01	D	3
<i>FSR-08</i>	SG-01	H1-01	D	3
<i>FSR-09</i>	SG-03	H1-04	D	3
<i>FSR-10</i>	SG-03	H1-04	D	3
<i>FSR-11</i>	SG-03	H1-04	D	3
<i>FSR-12</i>	SG-03	H1-04	D	3
<i>FSR-13</i>	SG-07	H4-02	B	2
<i>FSR-14</i>	SG-07	H4-02	B	2
<i>FSR-15</i>	SG-07	H4-02	B	2
<i>FSR-16</i>	SG-07	H4-02	B	2
<i>FSR-16A</i>	SG-01 / SG-07	H1-01, H2-01, H4-02	D	3
<i>FSR-17</i>	SG-06	H4-01	C	2
<i>FSR-18</i>	SG-06	H4-01	C	2
<i>FSR-19</i>	SG-06	H4-01	A	1
<i>FSR-20</i>	SG-01	H1-01	D	3
<i>FSR-21</i>	SG-06	H4-01	A	1
<i>FSR-22</i>	SG-01 / SG-06	H1-01, H2-01, H4-01	D	3

11. Verification and Validation Requirements

The laboratory implementation shall include a formal Verification and Validation (V&V) protocol.

Each test case shall document:

- Test identifier
- Sensor potentiometer settings
- Expected ECU response
- Actual CAN output
- CRC verification status

- Parallel readback verification result
- Pass/fail status
- Fault injection scenario
- Timestamp and operator information

The verification process shall validate:

- Crash detection functionality
- Non-deployment conditions
- Redundant channel agreement
- Communication integrity
- Diagnostic fault handling
- Safe-state transitions

12. References and Normative Standards

- ISO 26262:2018 — Road Vehicles — Functional Safety
- IEC 61508:2010 — Functional Safety of E/E/PE Systems
- ISO 21448 — SOTIF
- ISO 16750-3 — Environmental Conditions and Testing
- CISPR 25 — EMC Requirements
- ISO 11452 — Electrical Disturbance Testing
- ISO 14229 — Unified Diagnostic Services (UDS)
- ISO 11898 — Road Vehicles — Controller Area Network (CAN)
- SAE J211 — Impact Test Instrumentation
- FMVSS 208 — Occupant Crash Protection

13. Conclusion

The Airbag ECU laboratory implementation is classified as:

- ASIL D according to ISO 26262
- SIL 3 according to IEC 61508

The system therefore requires:

- Redundant dual-channel architecture
- Deterministic real-time execution
- CRC-protected CAN communication
- Continuous diagnostics
- Safe-state handling
- Fault-tolerant deployment logic
- Verified communication integrity
- Hardware-enforced deployment inhibit independent of software execution

The functional safety requirements defined in this specification establish the mandatory safety behavior necessary to ensure safe operation of the Airbag ECU prototype under both normal and fault conditions.

13.1 Use Case Screening Notes

The following operational use cases identified during the HARA process were screened out of detailed functional safety requirement derivation:

Use Case	Screening Justification
UC-10 Vehicle Wash / Cleaning	No crash detection or deployment scenario exists during stationary maintenance and cleaning activities. Residual risk is considered negligible under normal operating conditions.
UC-12 End-of-Life Vehicle Scrapping	End-of-life disassembly and decommissioning procedures are controlled through workshop safety procedures and are outside the operational scope of the laboratory ECU implementation.

14. Revision History

Revision	Date	Author	Change Description
1.0	2026-05-11	Systems Safety Engineering Team	Initial release of Functional Safety Requirements Specification derived from HARA and laboratory architecture
2.0	2026-05-11	Functional Safety Engineering Team	Added redundant 2oo2 architecture, CRC-protected CAN communication, timing constraints, and diagnostic safety requirements
2.1	2026-05-11	Functional Safety Engineering Team	Added safety rationales, CAN frame definition, memory integrity requirements, hardware inhibit constraints, and ASIL D communication integrity clarification

15. Approval and Sign-Off

Role	Name	Signature	Date
Functional Safety Engineer			
Software Safety Engineer			
Hardware Safety Engineer			
Project Supervisor			
Verification & Validation Lead			